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Институт №8 “Компьютерные науки и прикладная математика”
Кафедра №806 “Вычислительная математика и программирование”

Лабораторная работа №8 по курсу
«Операционные системы»

Группа: М8О-215Б-23

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Оценка:

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Постановка задачи

Вариант -

Цель работы: Приобретение практических навыков диагностики работы программного обеспечения.

Задание: при выполнении лабораторных работ по курсу ОС необходимо продемонстрировать ключевые системные вызовы, которые в них используются и то, что их использование соответствует варианту ЛР. По итогам выполнения всех лабораторных работ отчет по данной ЛР должен содержать краткую сводку по исследованию написанных программ.

Общий метод и алгоритм решения

Для демонстрации системных вызовов будем использовать утилиту `strace` в операционной системе `linux`.

`Strace` — это диагностическая, отладочная и обучающая утилита пользовательского пространства для `Linux`. Она используется для мониторинга и вмешательства во взаимодействия между процессами и ядром `Linux`, включая системные вызовы, доставку сигналов и изменение состояния процесса. Работа `strace` возможна благодаря функции ядра, известной как `ptrace`. Наиболее часто он используется для запуска программы и вывода списка системных вызовов, которые она выполняет. Это особенно полезно, когда программа часто аварийно завершает работу или ведет себя непредсказуемо; например, `strace` может показать, что программа пытается получить доступ к несуществующему или недоступному файлу.

Еще одно полезное применение `strace` — это использование флага `-p` для присоединения к уже запущенному процессу. Это может быть полезно, если процесс перестал отвечать, и `strace` может помочь определить, что процесс заблокирован при попытке установить сетевое соединение.

Кроме того, `strace` предоставляет множество других возможностей:

- **Фильтрация системных вызовов:** Вы можете указать, какие системные вызовы следует отслеживать, используя параметр `-e trace=`. Это можно сделать по имени (например, `clone`, `fork`, `vfork`), с использованием предопределенных групп (например, `%ipc` или `%file`) или с помощью регулярных выражений (начиная с версии `strace 4.17`, например, `-e trace=/clock_.*`).
- **Трассировка путей:** Вы можете указать список путей для трассировки, используя параметр `-P` (например, `-P /etc/ld.so.cache`).
- **Выгрузка ввода-вывода:** Вы можете указать список файловых дескрипторов для выгрузки ввода-вывода, используя параметры `-e read=` и `-e write=`.
- **Подсчет времени выполнения:** `Strace` позволяет подсчитывать время выполнения системных вызовов и их количество, используя параметры `-T`, `-s`, `-C` и `-w`. Параметр `-U` позволяет выводить дополнительную информацию, такую как минимальное и максимальное время выполнения системного вызова.
- **Временные метки:** `Strace` может печатать относительные или абсолютные временные метки, используя опции `-t` и `-r`.
- **Изменение системных вызовов:** `Strace` позволяет изменять выполняемые системные вызовы, включая изменение возвращаемого значения и кода ошибки, внедрение сигналов, задержек и изменение данных, на которые указывают аргументы системных вызовов. Это достигается с помощью опции `-e inject=syscall specification:tampering specification:retval=:error=:signal=:delay_enter=:delay_exit=:poke_enter=:poke_exit=`.
- **Информация о дескрипторах файлов:** `Strace` может извлекать информацию о дескрипторах файлов, включая сокеты, используя опцию `-u`. Опция `-uu` предоставляет дополнительную информацию, такую как адреса конечных точек для сокетов, пути и основные/дополнительные номера устройств для файлов.
- **Трассировка стека:** `Strace` может печатать трассировки стека, включая разборку символов (начиная с версии `strace 4.21`, используя опцию `-k`).

`Strace` также поддерживает множество других опций и параметров, которые делают его мощным инструментом

Лабораторная работа №1

Strace:

```

root@fa7703109f3c:/workspaces/OS_labs/lab1/src# strace -f ./parent
execve("./parent", ["/parent"], 0x7ffe2ce2eb88 /* 30 vars */) = 0
brk(NULL)                                = 0x1d36000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7f8a0de99000
access("/etc/ld.so.preload", R_OK)       = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=25258, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 25258, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f8a0de92000
close(3)                                 = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20t\2\0\0\0\0\0"...
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"...
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=1922136, ...}, AT_EMPTY_PATH) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"...
mmap(NULL, 1970000, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f8a0dcb1000
mmap(0x7f8a0dcd7000, 1396736, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x26000) = 0x7f8a0dcd7000
mmap(0x7f8a0de2c000, 339968, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3,
0x17b000) = 0x7f8a0de2c000
mmap(0x7f8a0de7f000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x1ce000) = 0x7f8a0de7f000
mmap(0x7f8a0de85000, 53072, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,
-1, 0) = 0x7f8a0de85000
close(3)                                 = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7f8a0dcae000
arch_prctl(ARCH_SET_FS, 0x7f8a0dcae740) = 0
set_tid_address(0x7f8a0dcae10)           = 7662
set_robust_list(0x7f8a0dcae20, 24)       = 0
rseq(0x7f8a0dc060, 0x20, 0, 0x53053053) = 0
mprotect(0x7f8a0de7f000, 16384, PROT_READ) = 0
mprotect(0x403000, 4096, PROT_READ)      = 0
mprotect(0x7f8a0dec000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7f8a0de92000, 25258)            = 0
pipe2([3, 4], 0)                         = 0
pipe2([5, 6], 0)                         = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD,
Process 7663 attached
<unfinished ...>
[pid 7663] set_robust_list(0x7f8a0dcae20, 24 <unfinished ...>
[pid 7662] <... clone resumed>, child_tidptr=0x7f8a0dcae10) = 7663
[pid 7663] <... set_robust_list resumed>) = 0
[pid 7662] close(3)                      = 0
[pid 7663] close(4 <unfinished ...>

```

```

[pid 7662] close(6 <unfinished ...>
[pid 7663] <... close resumed>) = 0
[pid 7662] <... close resumed>) = 0
[pid 7662] newfstatat(1, "", <unfinished ...>
[pid 7663] close(5 <unfinished ...>
[pid 7662] <... newfstatat resumed>{st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0),
..., AT_EMPTY_PATH) = 0
[pid 7663] <... close resumed>) = 0
[pid 7662] getrandom( <unfinished ...>
[pid 7663] dup2(3, 0 <unfinished ...>
[pid 7662] <... getrandom resumed>"\x97\xd4\x01\x83\x78\x26\x8d\x63", 8,
GRND_NONBLOCK) = 8
[pid 7663] <... dup2 resumed>) = 0
[pid 7662] brk(NULL <unfinished ...>
[pid 7663] dup2(6, 1 <unfinished ...>
[pid 7662] <... brk resumed>) = 0x1d36000
[pid 7663] <... dup2 resumed>) = 1
[pid 7662] brk(0x1d57000) = 0x1d57000
[pid 7663] execve("./child", ["child"], 0x7ffec332f4a8 /* 30 vars */ <unfinished ...>
[pid 7662] write(1, "\320\222\320\262\320\265\320\264\320\270\321\202\320\265

\320\264\320\260\320\275\320\275\321\213\320\265 \320\262 \321"... , 94Введите данные в
формате: «число число число<endline>».
) = 94
[pid 7662] newfstatat(0, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...},
AT_EMPTY_PATH) = 0
[pid 7662] read(0, <unfinished ...>
[pid 7663] <... execve resumed>) = 0
[pid 7663] brk(NULL) = 0x1e6e000
[pid 7663] mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7fe4e3c66000
[pid 7663] access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
[pid 7663] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 4
[pid 7663] newfstatat(4, "", {st_mode=S_IFREG|0644, st_size=25258, ...},
AT_EMPTY_PATH) = 0
[pid 7663] mmap(NULL, 25258, PROT_READ, MAP_PRIVATE, 4, 0) = 0x7fe4e3c5f000
[pid 7663] close(4) = 0
[pid 7663] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 4
[pid 7663] read(4,
"\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20t\2\0\0\0\0\0"... , 832) = 832
[pid 7663] pread64(4,
"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0\0"... , 784, 64) = 784
[pid 7663] newfstatat(4, "", {st_mode=S_IFREG|0755, st_size=1922136, ...},
AT_EMPTY_PATH) = 0
[pid 7663] pread64(4,
"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0\0"... , 784, 64) = 784
[pid 7663] mmap(NULL, 1970000, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 4, 0) =
0x7fe4e3a7e000
[pid 7663] mmap(0x7fe4e3aa4000, 1396736, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 4, 0x26000) = 0x7fe4e3aa4000

```

```

[pid 7663] mmap(0x7fe4e3bf9000, 339968, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 4, 0x17b000) = 0x7fe4e3bf9000
[pid 7663] mmap(0x7fe4e3c4c000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 4, 0x1ce000) = 0x7fe4e3c4c000
[pid 7663] mmap(0x7fe4e3c52000, 53072, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7fe4e3c52000
[pid 7663] close(4) = 0
[pid 7663] mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7fe4e3a7b000
[pid 7663] arch_prctl(ARCH_SET_FS, 0x7fe4e3a7b740) = 0
[pid 7663] set_tid_address(0x7fe4e3a7ba10) = 7663
[pid 7663] set_robust_list(0x7fe4e3a7ba20, 24) = 0
[pid 7663] rseq(0x7fe4e3a7c060, 0x20, 0, 0x53053053) = 0
[pid 7663] mprotect(0x7fe4e3c4c000, 16384, PROT_READ) = 0
[pid 7663] mprotect(0x403000, 4096, PROT_READ) = 0
[pid 7663] mprotect(0x7fe4e3c98000, 8192, PROT_READ) = 0
[pid 7663] prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
[pid 7663] munmap(0x7fe4e3c5f000, 25258) = 0
[pid 7663] read(0, 1 2 3 4
<unfinished ...>
[pid 7662] <... read resumed>"1 2 3 4\n", 1024) = 8
[pid 7662] write(4,
"\4\0\0\0\1\0\0\0\2\0\0\0\3\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 400) = 400
[pid 7663] <... read
400 resumed>"\4\0\0\0\1\0\0\0\2\0\0\0\3\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 400) =
[pid 7662] wait4(7663, <unfinished ...>
[pid 7663] openat(AT_FDCWD, "result.txt", O_WRONLY|O_CREAT|O_TRUNC, 0644) = 4
[pid 7663] write(4, "10\n", 3) = 3
[pid 7663] close(4) = 0
[pid 7663] exit_group(0) = ?
[pid 7663] +++ exited with 0 +++
<... wait4 resumed>[{WIFEXITED(s) && WEXITSTATUS(s) == 0}], 0, NULL) = 7663
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=7663, si_uid=0, si_status=0,
si_etime=0, si_stime=0} ---
close(4) = 0
close(5) = 0
write(1, "\320\237\321\200\320\276\320\263\321\200\320\260\320\274\320\274\320\260
\320\267\320\260\320\262\320\265\321\200\321\210\320"..., 71Программа завершила работу
без ошибок
) = 71
exit_group(0) = ?
+++ exited with 0 +++

```

Лабораторная работа №2

Strace:

```
root@fa7703109f3c:/workspaces/OS_labs/lab2/src# strace -f ./main.o 8
```

```
execve("./main.o", [ "./main.o", "8"], 0x7ffc37a40ab0 /* 30 vars */) = 0
```

```

brk(NULL) = 0x1448000

mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7fae2ce3e000

access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)

openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3

newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=25258, ...}, AT_EMPTY_PATH) = 0

mmap(NULL, 25258, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fae2ce37000

close(3) = 0

openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3

read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"... , 832) = 832

newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=907784, ...}, AT_EMPTY_PATH) = 0

mmap(NULL, 909560, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fae2cd58000

mmap(0x7fae2cd68000, 471040, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x10000) = 0x7fae2cd68000

mmap(0x7fae2cd8b000, 368640, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x83000)
= 0x7fae2cd8b000

mmap(0x7fae2ce35000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3,
0xdc000) = 0x7fae2ce35000

close(3) = 0

openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3

read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"... , 832) = 832

pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... , 784, 64)
= 784

newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=1922136, ...}, AT_EMPTY_PATH) = 0

pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... , 784, 64)
= 784

mmap(NULL, 1970000, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fae2cb77000

mmap(0x7fae2cb9d000, 1396736, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x26000) = 0x7fae2cb9d000

mmap(0x7fae2ccf2000, 339968, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3,
0x17b000) = 0x7fae2ccf2000

mmap(0x7fae2cd45000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x1ce000) = 0x7fae2cd45000

mmap(0x7fae2cd4b000, 53072, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,
-1, 0) = 0x7fae2cd4b000

close(3) = 0

mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7fae2cb74000

arch_prctl(ARCH_SET_FS, 0x7fae2cb74740) = 0

```

```

set_tid_address(0x7fae2cb74a10)      = 88342

set_robust_list(0x7fae2cb74a20, 24)   = 0

rseq(0x7fae2cb75060, 0x20, 0, 0x53053053) = 0

mprotect(0x7fae2cd45000, 16384, PROT_READ) = 0

mprotect(0x7fae2ce35000, 4096, PROT_READ) = 0

mprotect(0x403000, 4096, PROT_READ)   = 0

mprotect(0x7fae2ce70000, 8192, PROT_READ) = 0

prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0

munmap(0x7fae2ce37000, 25258)         = 0

newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}, AT_EMPTY_PATH)
= 0

getrandom("\xcf\x50\x7f\x0b\x89\x8a\xf5\xec", 8, GRND_NONBLOCK) = 8

brk(NULL)                             = 0x1448000

brk(0x1469000)                        = 0x1469000

newfstatat(0, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}, AT_EMPTY_PATH)
= 0

write(1, "Enter the number of elements in "..., 43Enter the number of elements in the
array: ) = 43

read(0, 1000

"1000\n", 1024)                       = 5

rt_sigaction(SIGRT_1, {sa_handler=0x7fae2cbfd6a0, sa_mask=[],
sa_flags=SA_RESTORER|SA_ONSTACK|SA_RESTART|SA_SIGINFO, sa_restorer=0x7fae2cbb3050}, NULL, 8)
= 0

rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0

mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) =
0x7fae2c373000

mprotect(0x7fae2c374000, 8388608, PROT_READ|PROT_WRITE) = 0

rt_sigprocmask(SIG_BLOCK, ~[], [], 8)   = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLO
NE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7fae2cb73990,
parent_tid=0x7fae2cb73990, exit_signal=0, stack=0x7fae2c373000, stack_size=0x7fff80,
tls=0x7fae2cb736c0}strace: Process 88370 attached

=> {parent_tid=[88370]}, 88) = 88370

[pid 88370] rseq(0x7fae2cb73fe0, 0x20, 0, 0x53053053 <unfinished ...>

[pid 88342] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 88370] <... rseq resumed>)         = 0

[pid 88342] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 88370] set_robust_list(0x7fae2cb739a0, 24 <unfinished ...>

```

```
[pid 88342] mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0
<unfinished ...>
```

```
[pid 88370] <... set_robust_list resumed>) = 0
```

```
[pid 88342] <... mmap resumed>) = 0x7fae2bb72000
```

```
[pid 88370] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
```

```
[pid 88342] mprotect(0x7fae2bb73000, 8388608, PROT_READ|PROT_WRITE <unfinished ...>
```

```
[pid 88370] <... rt_sigprocmask resumed>NULL, 8) = 0
```

```
[pid 88342] <... mprotect resumed>) = 0
```

```
[pid 88342] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
```

```
[pid 88342]
```

```
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SE
TTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7fae2c372990,
parent_tid=0x7fae2c372990, exit_signal=0, stack=0x7fae2bb72000, stack_size=0x7fff80,
tls=0x7fae2c3726c0}strace: Process 88371 attached
```

```
=> {parent_tid=[88371]}, 88) = 88371
```

```
[pid 88371] rseq(0x7fae2c372fe0, 0x20, 0, 0x53053053 <unfinished ...>
```

```
[pid 88342] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
```

```
[pid 88371] <... rseq resumed>) = 0
```

```
[pid 88342] <... rt_sigprocmask resumed>NULL, 8) = 0
```

```
[pid 88371] set_robust_list(0x7fae2c3729a0, 24 <unfinished ...>
```

```
[pid 88342] mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0
<unfinished ...>
```

```
[pid 88371] <... set_robust_list resumed>) = 0
```

```
[pid 88342] <... mmap resumed>) = 0x7fae2b371000
```

```
[pid 88371] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
```

```
[pid 88342] mprotect(0x7fae2b372000, 8388608, PROT_READ|PROT_WRITE <unfinished ...>
```

```
[pid 88371] <... rt_sigprocmask resumed>NULL, 8) = 0
```

```
[pid 88342] <... mprotect resumed>) = 0
```

```
[pid 88342] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
```

```
[pid 88342]
```

```
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SE
TTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7fae2bb71990,
parent_tid=0x7fae2bb71990, exit_signal=0, stack=0x7fae2b371000, stack_size=0x7fff80,
tls=0x7fae2bb716c0}strace: Process 88372 attached
```

```
<unfinished ...>
```

```
[pid 88372] rseq(0x7fae2bb71fe0, 0x20, 0, 0x53053053 <unfinished ...>
```

```
[pid 88342] <... clone3 resumed> => {parent_tid=[88372]}, 88) = 88372
```

```
[pid 88372] <... rseq resumed>) = 0
```



```

[pid 88342] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 88372] set_robust_list(0x7fae2bb719a0, 24 <unfinished ...>

[pid 88342] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 88372] <... set_robust_list resumed>) = 0

[pid 88342] mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0
<unfinished ...>

[pid 88372] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 88342] <... mmap resumed>) = 0x7fae2ab70000

[pid 88372] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 88342] mprotect(0x7fae2ab71000, 8388608, PROT_READ|PROT_WRITE) = 0

[pid 88342] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

[pid 88342]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SE
TTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7fae2b370990,
parent_tid=0x7fae2b370990, exit_signal=0, stack=0x7fae2ab70000, stack_size=0x7fff80,
tls=0x7fae2b3706c0}strace: Process 88373 attached

=> {parent_tid=[88373]}, 88) = 88373

[pid 88373] rseq(0x7fae2b370fe0, 0x20, 0, 0x53053053 <unfinished ...>

[pid 88342] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 88373] <... rseq resumed>) = 0

[pid 88342] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 88373] set_robust_list(0x7fae2b3709a0, 24 <unfinished ...>

[pid 88342] mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0
<unfinished ...>

[pid 88373] <... set_robust_list resumed>) = 0

[pid 88342] <... mmap resumed>) = 0x7fae2a36f000

[pid 88373] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 88342] mprotect(0x7fae2a370000, 8388608, PROT_READ|PROT_WRITE <unfinished ...>

[pid 88373] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 88342] <... mprotect resumed>) = 0

[pid 88342] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

[pid 88342]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SE
TTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7fae2ab6f990,
parent_tid=0x7fae2ab6f990, exit_signal=0, stack=0x7fae2a36f000, stack_size=0x7fff80,
tls=0x7fae2ab6f6c0}strace: Process 88374 attached

=> {parent_tid=[88374]}, 88) = 88374

[pid 88342] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

```

```

[pid 88374] rseq(0x7fae2ab6ffe0, 0x20, 0, 0x53053053 <unfinished ...>

[pid 88342] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 88374] <... rseq resumed>) = 0

[pid 88342] mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0
<unfinished ...>

[pid 88374] set_robust_list(0x7fae2ab6f9a0, 24 <unfinished ...>

[pid 88342] <... mmap resumed>) = 0x7fae29b6e000

[pid 88342] mprotect(0x7fae29b6f000, 8388608, PROT_READ|PROT_WRITE <unfinished ...>

[pid 88374] <... set_robust_list resumed>) = 0

[pid 88342] <... mprotect resumed>) = 0

[pid 88342] rt_sigprocmask(SIG_BLOCK, ~[], <unfinished ...>

[pid 88374] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 88342] <... rt_sigprocmask resumed>[], 8) = 0

[pid 88374] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 88342]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SE
TTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7fae2a36e990,
parent_tid=0x7fae2a36e990, exit_signal=0, stack=0x7fae29b6e000, stack_size=0x7fff80,
tls=0x7fae2a36e6c0} => {parent_tid=[88375]}, 88) = 88375

[pid 88342] rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0

[pid 88342] mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) =
0x7fae2936d000

[pid 88342] mprotect(0x7fae2936e000, 8388608, PROT_READ|PROT_WRITE) = 0

[pid 88342] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

[pid 88342]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SE
TTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7fae29b6d990,
parent_tid=0x7fae29b6d990, exit_signal=0, stack=0x7fae2936d000, stack_size=0x7fff80,
tls=0x7fae29b6d6c0} => {parent_tid=[88376]}, 88) = 88376

[pid 88342] rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0

[pid 88342] mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) =
0x7fae28b6c000

[pid 88342] mprotect(0x7fae28b6d000, 8388608, PROT_READ|PROT_WRITE) = 0

[pid 88342] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

[pid 88342]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SE
TTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7fae2936c990,
parent_tid=0x7fae2936c990, exit_signal=0, stack=0x7fae28b6c000, stack_size=0x7fff80,
tls=0x7fae2936c6c0}strace: Process 88377 attached

=> {parent_tid=[88377]}, 88) = 88377

```

```

[pid 88377] rseq(0x7fae2936cfe0, 0x20, 0, 0x53053053 <unfinished ...>
[pid 88342] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 88377] <... rseq resumed>) = 0
[pid 88342] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 88377] set_robust_list(0x7fae2936c9a0, 24 <unfinished ...>
[pid 88342] futex(0x7fae2cb73990, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 88370, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>
[pid 88377] <... set_robust_list resumed>) = 0
[pid 88377] rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0
strace: Process 88376 attached
strace: Process 88375 attached
[pid 88376] rseq(0x7fae29b6dfe0, 0x20, 0, 0x53053053 <unfinished ...>
[pid 88375] rseq(0x7fae2a36efe0, 0x20, 0, 0x53053053 <unfinished ...>
[pid 88376] <... rseq resumed>) = 0
[pid 88375] <... rseq resumed>) = 0
[pid 88376] set_robust_list(0x7fae29b6d9a0, 24 <unfinished ...>
[pid 88375] set_robust_list(0x7fae2a36e9a0, 24 <unfinished ...>
[pid 88376] <... set_robust_list resumed>) = 0
[pid 88375] <... set_robust_list resumed>) = 0
[pid 88376] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 88375] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 88376] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 88375] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 88376] mmap(NULL, 134217728, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_NORESERVE,
-1, 0) = 0x7fae20b6c000
[pid 88376] munmap(0x7fae20b6c000, 55132160) = 0
[pid 88376] munmap(0x7fae28000000, 11976704) = 0
[pid 88376] mprotect(0x7fae24000000, 135168, PROT_READ|PROT_WRITE) = 0
[pid 88376] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
[pid 88376] newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=25258, ...}, AT_EMPTY_PATH)
= 0
[pid 88376] mmap(NULL, 25258, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fae2ce37000
[pid 88376] close(3) = 0
[pid 88376] openat(AT_FDCWD, "/usr/local/lib64/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
[pid 88376] read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"...
832) = 832

```

```

[pid 88376] newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=906528, ...},
AT_EMPTY_PATH) = 0

[pid 88376] mmap(NULL, 181160, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7fae28b3f000

[pid 88376] mmap(0x7fae28b43000, 143360, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x7fae28b43000

[pid 88376] mmap(0x7fae28b66000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x27000) = 0x7fae28b66000

[pid 88376] mmap(0x7fae28b6a000, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2b000) = 0x7fae28b6a000

[pid 88376] close(3) = 0

[pid 88376] mprotect(0x7fae28b6a000, 4096, PROT_READ) = 0

[pid 88376] munmap(0x7fae2ce37000, 25258) = 0

[pid 88376] futex(0x7fae28b6b230, FUTEX_WAKE_PRIVATE, 2147483647) = 0

[pid 88376] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0

[pid 88376] madvise(0x7fae2936d000, 8368128, MADV_DONTNEED) = 0

[pid 88376] exit(0) = ?

[pid 88376] +++ exited with 0 +++

[pid 88371] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0

[pid 88371] madvise(0x7fae2bb72000, 8368128, MADV_DONTNEED) = 0

[pid 88371] exit(0) = ?

[pid 88371] +++ exited with 0 +++

[pid 88375] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0

[pid 88375] madvise(0x7fae29b6e000, 8368128, MADV_DONTNEED) = 0

[pid 88375] exit(0) = ?

[pid 88375] +++ exited with 0 +++

[pid 88377] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0

[pid 88377] madvise(0x7fae28b6c000, 8368128, MADV_DONTNEED) = 0

[pid 88377] exit(0) = ?

[pid 88377] +++ exited with 0 +++

[pid 88372] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0

[pid 88372] madvise(0x7fae2b371000, 8368128, MADV_DONTNEED) = 0

[pid 88372] exit(0) = ?

[pid 88372] +++ exited with 0 +++

[pid 88374] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0

[pid 88374] madvise(0x7fae2a36f000, 8368128, MADV_DONTNEED) = 0

```

```
[pid 88374] exit(0) = ?

[pid 88374] +++ exited with 0 +++

[pid 88373] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0

[pid 88373] madvise(0x7fae2ab70000, 8368128, MADV_DONTNEED) = 0

[pid 88373] exit(0) = ?

[pid 88373] +++ exited with 0 +++

[pid 88370] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0

[pid 88370] madvise(0x7fae2c373000, 8368128, MADV_DONTNEED) = 0

[pid 88370] exit(0) = ?

[pid 88342] <... futex resumed> = 0

[pid 88370] +++ exited with 0 +++

munmap(0x7fae2c373000, 8392704) = 0

munmap(0x7fae2bb72000, 8392704) = 0

munmap(0x7fae2b371000, 8392704) = 0

munmap(0x7fae2ab70000, 8392704) = 0

write(1, "{38.00, 80.00, 51.00} - Point 0\n", 32{38.00, 80.00, 51.00} - Point 0
) = 32

write(1, "{46.00, 5.00, 44.00} - Point 1\n", 31{46.00, 5.00, 44.00} - Point 1
) = 31

write(1, "{90.00, 2.00, 69.00} - Point 2\n", 31{90.00, 2.00, 69.00} - Point 2
) = 31

write(1, "1909.519311 - Max area\n", 231909.519311 - Max area
) = 23

write(1, "Time taken for calculating: 1.26"... , 45Time taken for calculating: 1.269246
seconds
) = 45

write(1, "Number of threads used: 8\n", 26Number of threads used: 8
) = 26

lseek(0, -1, SEEK_CUR) = -1 ESPIPE (Illegal seek)

exit_group(0) = ?

+++ exited with 0 +++
```

Лабораторная работа №3

Strace:

```
root@fa7703109f3c:/workspaces/OS_labs/lab3/src# strace -f ./parent.o

execve("./parent.o", [ "./parent.o" ], 0x7ffdb9123a48 /* 30 vars */) = 0

brk(NULL)                               = 0xaf9000

mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7f04bfaf0000

access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)

openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3

newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=25258, ...}, AT_EMPTY_PATH) = 0

mmap(NULL, 25258, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f04bfae9000

close(3)                                = 0

openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3

read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20t\2\0\0\0\0\0"... , 832) = 832

pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... , 784, 64)
= 784

newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=1922136, ...}, AT_EMPTY_PATH) = 0

pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... , 784, 64)
= 784

mmap(NULL, 1970000, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f04bf908000

mmap(0x7f04bf92e000, 1396736, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x26000) = 0x7f04bf92e000

mmap(0x7f04bfa83000, 339968, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3,
0x17b000) = 0x7f04bfa83000

mmap(0x7f04bfad6000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x1ce000) = 0x7f04bfad6000

mmap(0x7f04bfadc000, 53072, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,
-1, 0) = 0x7f04bfadc000

close(3)                                = 0

mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7f04bf905000

arch_prctl(ARCH_SET_FS, 0x7f04bf905740) = 0

set_tid_address(0x7f04bf905a10)         = 87501

set_robust_list(0x7f04bf905a20, 24)     = 0

rseq(0x7f04bf906060, 0x20, 0, 0x53053053) = 0

mprotect(0x7f04bfad6000, 16384, PROT_READ) = 0

mprotect(0x403000, 4096, PROT_READ)     = 0
```

```

mprotect(0x7f04bfb22000, 8192, PROT_READ) = 0

prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0

munmap(0x7f04bfae9000, 25258) = 0

openat(AT_FDCWD, "/my_shared_memory", O_RDWR|O_CREAT, 0666) = 3

ftruncate(3, 656) = 0

mmap(NULL, 656, PROT_READ|PROT_WRITE, MAP_SHARED, 3, 0) = 0x7f04bfaef000

close(3) = 0

openat(AT_FDCWD, "/dev/shm/sem.sem_parent", O_RDWR|O_NOFOLLOW) = -1 ENOENT (No such file
or directory)

getrandom("\x17\x2f\x13\x56\x16\xe4\x54\x90", 8, GRND_NONBLOCK) = 8

newfstatat(AT_FDCWD, "/dev/shm/sem.ZAWF15", 0x7ffe8b5db410, AT_SYMLINK_NOFOLLOW) = -1
ENOENT (No such file or directory)

openat(AT_FDCWD, "/dev/shm/sem.ZAWF15", O_RDWR|O_CREAT|O_EXCL, 0666) = 3

write(3, "\0\0\0\0\0\0\0\0\200\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0", 32) = 32

mmap(NULL, 32, PROT_READ|PROT_WRITE, MAP_SHARED, 3, 0) = 0x7f04bfaee000

link("/dev/shm/sem.ZAWF15", "/dev/shm/sem.sem_parent") = 0

newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=32, ...}, AT_EMPTY_PATH) = 0

getrandom("\x00\x3e\xae\xf5\x64\xd3\x32\x78", 8, GRND_NONBLOCK) = 8

brk(NULL) = 0xaf9000

brk(0xb1a000) = 0xb1a000

unlink("/dev/shm/sem.ZAWF15") = 0

close(3) = 0

openat(AT_FDCWD, "/dev/shm/sem.sem_child", O_RDWR|O_NOFOLLOW) = -1 ENOENT (No such file
or directory)

getrandom("\x1c\xb8\x82\xa5\x01\x13\x3d\x70", 8, GRND_NONBLOCK) = 8

newfstatat(AT_FDCWD, "/dev/shm/sem.uM3BBt", 0x7ffe8b5db410, AT_SYMLINK_NOFOLLOW) = -1
ENOENT (No such file or directory)

openat(AT_FDCWD, "/dev/shm/sem.uM3BBt", O_RDWR|O_CREAT|O_EXCL, 0666) = 3

write(3, "\0\0\0\0\0\0\0\0\200\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0", 32) = 32

mmap(NULL, 32, PROT_READ|PROT_WRITE, MAP_SHARED, 3, 0) = 0x7f04bfaed000

link("/dev/shm/sem.uM3BBt", "/dev/shm/sem.sem_child") = 0

newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=32, ...}, AT_EMPTY_PATH) = 0

unlink("/dev/shm/sem.uM3BBt") = 0

close(3) = 0

clone(child_stack=NULL, flags=CLONE_CHILD_CLEARPID|CLONE_CHILD_SETTID|SIGCHLD, strace:
Process 87504 attached

```

```

, child_tidptr=0x7f04bf905a10) = 87504

[pid 87504] set_robust_list(0x7f04bf905a20, 24 <unfinished ...>

[pid 87501] newfstatat(1, "", <unfinished ...>

[pid 87504] <... set_robust_list resumed>) = 0

[pid 87501] <... newfstatat resumed>{st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0),
...}, AT_EMPTY_PATH) = 0

[pid 87501] write(1, "\320\222\320\262\320\265\320\264\320\270\321\202\320\265
\320\264\320\260\320\275\320\275\321\213\320\265 \320\262 \321"..., 94 <unfinished ...>

Введите данные в формате: «число число число<newline>».

[pid 87504] execve("./child.o", ["child"], 0x7ffe8b5dbe78 /* 30 vars */ <unfinished ...>

[pid 87501] <... write resumed>) = 94

[pid 87501] newfstatat(0, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...},
AT_EMPTY_PATH) = 0

[pid 87501] read(0, <unfinished ...>

[pid 87504] <... execve resumed>) = 0

[pid 87504] brk(NULL) = 0x11fd000

[pid 87504] mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7f5b4fa87000

[pid 87504] access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)

[pid 87504] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3

[pid 87504] newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=25258, ...}, AT_EMPTY_PATH)
= 0

[pid 87504] mmap(NULL, 25258, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f5b4fa80000

[pid 87504] close(3) = 0

[pid 87504] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3

[pid 87504] read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20t\2\0\0\0\0"...,
832) = 832

[pid 87504] pread64(3,
"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784

[pid 87504] newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=1922136, ...},
AT_EMPTY_PATH) = 0

[pid 87504] pread64(3,
"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784

[pid 87504] mmap(NULL, 1970000, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7f5b4f89f000

[pid 87504] mmap(0x7f5b4f8c5000, 1396736, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x26000) = 0x7f5b4f8c5000

[pid 87504] mmap(0x7f5b4fa1a000, 339968, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x17b000) = 0x7f5b4fa1a000

```



```

[pid 87504] mmap(0x7f5b4fa6d000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1ce000) = 0x7f5b4fa6d000

[pid 87504] mmap(0x7f5b4fa73000, 53072, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f5b4fa73000

[pid 87504] close(3) = 0

[pid 87504] mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7f5b4f89c000

[pid 87504] arch_prctl(ARCH_SET_FS, 0x7f5b4f89c740) = 0

[pid 87504] set_tid_address(0x7f5b4f89ca10) = 87504

[pid 87504] set_robust_list(0x7f5b4f89ca20, 24) = 0

[pid 87504] rseq(0x7f5b4f89d060, 0x20, 0, 0x53053053) = 0

[pid 87504] mprotect(0x7f5b4fa6d000, 16384, PROT_READ) = 0

[pid 87504] mprotect(0x403000, 4096, PROT_READ) = 0

[pid 87504] mprotect(0x7f5b4fab9000, 8192, PROT_READ) = 0

[pid 87504] prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0

[pid 87504] munmap(0x7f5b4fa80000, 25258) = 0

[pid 87504] openat(AT_FDCWD, "/my_shared_memory", O_RDWR) = 3

[pid 87504] mmap(NULL, 656, PROT_READ|PROT_WRITE, MAP_SHARED, 3, 0) = 0x7f5b4fa86000

[pid 87504] close(3) = 0

[pid 87504] openat(AT_FDCWD, "/dev/shm/sem.sem_parent", O_RDWR|O_NOFOLLOW) = 3

[pid 87504] newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=32, ...}, AT_EMPTY_PATH) =
0

[pid 87504] getrandom("\x48\x27\x2b\x02\xb6\x69\x51\x04", 8, GRND_NONBLOCK) = 8

[pid 87504] brk(NULL) = 0x11fd000

[pid 87504] brk(0x121e000) = 0x121e000

[pid 87504] mmap(NULL, 32, PROT_READ|PROT_WRITE, MAP_SHARED, 3, 0) = 0x7f5b4fa85000

[pid 87504] close(3) = 0

[pid 87504] openat(AT_FDCWD, "/dev/shm/sem.sem_child", O_RDWR|O_NOFOLLOW) = 3

[pid 87504] newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=32, ...}, AT_EMPTY_PATH) =
0

[pid 87504] mmap(NULL, 32, PROT_READ|PROT_WRITE, MAP_SHARED, 3, 0) = 0x7f5b4fa84000

[pid 87504] close(3) = 0

[pid 87504] futex(0x7f5b4fa84000, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>

[pid 87501] <... read resumed>0xaf9770, 1024) = ? ERESTARTSYS (To be restarted if
SA_RESTART is set)

```

```

[pid 87504] <... futex resumed>)          = ? ERESTARTSYS (To be restarted if SA_RESTART
is set)

[pid 87501] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---

[pid 87504] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---

[pid 87501] read(0, <unfinished ...>

[pid 87504] futex(0x7f5b4fa84000, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL,
FUTEX_BITSET_MATCH_ANY) = ? ERESTARTSYS (To be restarted if SA_RESTART is set)

[pid 87501] <... read resumed>0xaf9770, 1024) = ? ERESTARTSYS (To be restarted if
SA_RESTART is set)

[pid 87504] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---

[pid 87501] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---

[pid 87504] futex(0x7f5b4fa84000, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>

[pid 87501] read(0, 0xaf9770, 1024)      = ? ERESTARTSYS (To be restarted if SA_RESTART
is set)

[pid 87504] <... futex resumed>)          = ? ERESTARTSYS (To be restarted if SA_RESTART
is set)

[pid 87501] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---

[pid 87504] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---

[pid 87501] read(0, <unfinished ...>

[pid 87504] futex(0x7f5b4fa84000, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>

[pid 87501] <... read resumed>0xaf9770, 1024) = ? ERESTARTSYS (To be restarted if
SA_RESTART is set)

[pid 87504] <... futex resumed>)          = ? ERESTARTSYS (To be restarted if SA_RESTART
is set)

[pid 87501] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---

[pid 87504] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---

[pid 87501] read(0, <unfinished ...>

[pid 87504] futex(0x7f5b4fa84000, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>

[pid 87501] <... read resumed>0xaf9770, 1024) = ? ERESTARTSYS (To be restarted if
SA_RESTART is set)

[pid 87504] <... futex resumed>)          = ? ERESTARTSYS (To be restarted if SA_RESTART
is set)

[pid 87501] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---

[pid 87504] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---

[pid 87501] read(0, <unfinished ...>

```

```

[pid 87504] futex(0x7f5b4fa84000, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>

[pid 87501] <... read resumed>0xaf9770, 1024) = ? ERESTARTSYS (To be restarted if
SA_RESTART is set)

[pid 87504] <... futex resumed>          = ? ERESTARTSYS (To be restarted if SA_RESTART
is set)

[pid 87501] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---
[pid 87504] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---
[pid 87501] read(0, <unfinished ...>

[pid 87504] futex(0x7f5b4fa84000, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>

[pid 87501] <... read resumed>0xaf9770, 1024) = ? ERESTARTSYS (To be restarted if
SA_RESTART is set)

[pid 87504] <... futex resumed>          = ? ERESTARTSYS (To be restarted if SA_RESTART
is set)

[pid 87501] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---
[pid 87504] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---
[pid 87501] read(0, <unfinished ...>

[pid 87504] futex(0x7f5b4fa84000, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>

[pid 87501] <... read resumed>0xaf9770, 1024) = ? ERESTARTSYS (To be restarted if
SA_RESTART is set)

[pid 87504] <... futex resumed>          = ? ERESTARTSYS (To be restarted if SA_RESTART
is set)

[pid 87501] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---
[pid 87504] --- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---
[pid 87501] read(0, <unfinished ...>

[pid 87504] futex(0x7f5b4fa84000, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL,
FUTEX_BITSET_MATCH_ANY1 2 3 4 5

<unfinished ...>

```

```

[pid 87501] <... read resumed>"1 2 3 4 5\n", 1024) = 10
[pid 87501] futex(0x7f04bfaed000, FUTEX_WAKE, 1) = 1
[pid 87504] <... futex resumed>) = 0
[pid 87501] futex(0x7f04bfaee000, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>
[pid 87504] openat(AT_FDCWD, "result.txt", O_WRONLY|O_CREAT|O_TRUNC, 0666) = 3
[pid 87504] write(3, "15\n", 3) = 3
[pid 87504] close(3) = 0
[pid 87504] futex(0x7f5b4fa85000, FUTEX_WAKE, 1 <unfinished ...>
[pid 87501] <... futex resumed>) = 0
[pid 87504] <... futex resumed>) = 1
[pid 87501] write(1,
"\320\240\320\265\320\267\321\203\320\273\321\214\321\202\320\260\321\202:
\320\241\321\203\320\274\320\274\320\260 \320"... , 60 <unfinished ...>

Результат: Сумма записана в файл
[pid 87504] munmap(0x7f5b4fa86000, 656 <unfinished ...>
[pid 87501] <... write resumed>) = 60
[pid 87504] <... munmap resumed>) = 0
[pid 87501] wait4(87504, <unfinished ...>
[pid 87504] exit_group(0) = ?
[pid 87504] +++ exited with 0 +++
<... wait4 resumed>NULL, 0, NULL) = 87504
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=87504, si_uid=0, si_status=0,
si_utime=0, si_stime=1 /* 0.01 s */} ---
munmap(0x7f04bfaef000, 656) = 0
munmap(0x7f04bfaee000, 32) = 0
munmap(0x7f04bfaed000, 32) = 0
unlink("/dev/shm/sem.sem_parent") = 0
unlink("/dev/shm/sem.sem_child") = 0
exit_group(0) = ?
+++ exited with 0 +++

```

Лабораторная работа №4

Strace:

test_static.c

```

execve("/test_static", ["/test_static"], 0x7ffc3fb39040 /* 34 vars */) =
0 brk(NULL) = 0x558b99efb000
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffc24c602c0) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7ff2e41af000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/glibc-hwcaps/x86-64-v3/libImpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or
directory) openat(AT_FDCWD, "/glibc-hwcaps/x86-64-v2/libImpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or
directory) openat(AT_FDCWD, "/tls/x86_64/x86_64/libImpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or
directory) openat(AT_FDCWD, "/tls/x86_64/libImpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/tls/x86_64/libImpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/tls/libImpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/x86_64/x86_64/libImpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/x86_64/libImpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/x86_64/libImpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/libImpl1.so", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"... , 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0777, st_size=15648, ...}, AT_EMPTY_PATH) =
0 getcwd("/mnt/d/Projects/egorka_os/lab4/src", 128) = 35
mmap(NULL, 16432, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7ff2e41aa000
mmap(0x7ff2e41ab000, 4096, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1000) = 0x7ff2e41ab000
mmap(0x7ff2e41ac000, 4096, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x7ff2e41ac000
mmap(0x7ff2e41ad000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) =
0x7ff2e41ad000
close(3) = 0
openat(AT_FDCWD, "/glibc-hwcaps/x86-64-v3/libc.so.6", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or
directory) openat(AT_FDCWD, "/glibc-hwcaps/x86-64-v2/libc.so.6", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file
or directory) openat(AT_FDCWD, "/tls/x86_64/x86_64/libc.so.6", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or
directory) openat(AT_FDCWD, "/tls/x86_64/libc.so.6", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/tls/x86_64/libc.so.6", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/tls/libc.so.6", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/x86_64/x86_64/libc.so.6", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/x86_64/libc.so.6", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/x86_64/libc.so.6", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/libc.so.6", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=22187, ...}, AT_EMPTY_PATH) =
0 mmap(NULL, 22187, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7ff2e41a4000
close(3) =

```

```

openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3 read(3,
"\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
pread64(3, "\4\0\0\0\0\0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0"..., 48, 848) = 48
pread64(3, "\4\0\0\0\24\0\0\0\3\0\0\0GNU\0\1\7\357\204\3$\f\221\2039x\324\224\323\236S"..., 68, 896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7ff2e3f7b000
mprotect(0x7ff2e3fa3000, 2023424, PROT_NONE) = 0
mmap(0x7ff2e3fa3000, 1658880, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) =
0x7ff2e3fa3000
mmap(0x7ff2e4138000, 360448, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7ff2e4138000
mmap(0x7ff2e4191000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000)
= 0x7ff2e4191000
mmap(0x7ff2e4197000, 52816, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7ff2e4197000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=940560, ...}, AT_EMPTY_PATH) =
0
mmap(NULL, 942344, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7ff2e3e94000
mmap(0x7ff2e3ea2000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xe000) =
0x7ff2e3ea2000
mmap(0x7ff2e3f1e000, 372736, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x8a000) =
0x7ff2e3f1e000
mmap(0x7ff2e3f79000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xe4000) =
0x7ff2e3f79000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7ff2e3e91000
arch_prctl(ARCH_SET_FS, 0x7ff2e3e91740) =
0
set_tid_address(0x7ff2e3e91a10) = 57510
set_robust_list(0x7ff2e3e91a20, 24) = 0
rseq(0x7ff2e3e920e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7ff2e4191000, 16384, PROT_READ) = 0
mprotect(0x7ff2e3f79000, 4096, PROT_READ) = 0
mprotect(0x7ff2e41ad000, 4096, PROT_READ) = 0
mprotect(0x558b7addf000, 4096, PROT_READ) = 0
mprotect(0x7ff2e41e9000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) =
0
munmap(0x7ff2e41a4000, 22187) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x7), ...}, AT_EMPTY_PATH) = 0
getrandom("\x13\xf4\xf5\x10\xee\x37\xc5\xe3", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0x558b99efb000
brk(0x558b99f1c000) =
0x558b99f1c000
write(1, "Input program code:\n",
20) = 20
write(1, " 1 -> PrimeCount\n", 17) = 17
write(1, " 2 -> Pi\n", 9) = 9
write(1, "-1 -> Exit\n", 11) = 11
newfstatat(0, "", {st_mode=S_IFIFO|0600, st_size=0, ...}, AT_EMPTY_PATH) =
0
read(0, "\1\n0 \10\n2\n100\n-1\n", 4096) = 16
write(1, "Enter A and B: PrimeCount(0, 10)...., 37) = 37
write(1, "Input program code:\n", 20) = 20
write(1, " 1 -> PrimeCount\n", 17) = 17
write(1, " 2 -> Pi\n", 9) = 9
write(1, "-1 -> Exit\n", 11) = 11
write(1, "Enter K: Pi(100) = 3.131593\n", 28) = 28
write(1, "Input program code:\n", 20) = 20
write(1, " 1 -> PrimeCount\n", 17) = 17

```

```

write(1, " 2 -> Pi\n", 9)          = 9
write(1, "-1 -> Exit\n", 11)       = 11
lseek(0, -1, SEEK_CUR)              = -1 ESPIPE (Illegal
seek) exit_group(0)                = ?
+++ exited with 0 +++

```

test_dynamic.c:

```

execve("./test_dynamic", [".test_dynamic"], 0x7ffe7c042fc0 /* 34 vars */) =
0 brk(NULL)                          = 0x563b25b55000
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffcd595b920) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f84073e2000
access("/etc/ld.so.preload", R_OK)    = -1 ENOENT (No such file or
directory) openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=22187, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 22187, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f84073dc000
close(3)                             = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0"... , 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0\0@\0\0\0\0\0\0\0\0\0\0\0"... , 784, 64) = 784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"... , 48, 848) = 48
pread64(3, "\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"... , 68, 896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0\0@\0\0\0\0\0\0\0\0\0\0\0"... , 784, 64) = 784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f84071b3000
mprotect(0x7f84071db000, 2023424, PROT_NONE) = 0
mmap(0x7f84071db000, 1658880, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) =
0x7f84071db000
mmap(0x7f8407370000, 360448, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7f8407370000
mmap(0x7f84073c9000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) =
0x7f84073c9000
mmap(0x7f84073cf000, 52816, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) =
0x7f84073cf000
close(3)                             = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f84071b0000
arch_prctl(ARCH_SET_FS, 0x7f84071b0740) = 0
set_tid_address(0x7f84071b0a10)      =
62096
set_robust_list(0x7f84071b0a20, 24)  = 0
rseq(0x7f84071b10e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7f84073c9000, 16384, PROT_READ) = 0
mprotect(0x563b11e2d000, 4096, PROT_READ) = 0
mprotect(0x7f840741c000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) =
0 munmap(0x7f84073dc000, 22187)      = 0
getrandom("\xa2\x51\x02\x5f\x26\xb4\x3a\x43", 8, GRND_NONBLOCK) =
8 brk(NULL)                          = 0x563b25b55000
brk(0x563b25b76000)                  = 0x563b25b76000
openat(AT_FDCWD, "./libImpl1.so", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"... , 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0777, st_size=15648, ...}, AT_EMPTY_PATH) = 0
getcwd("/mnt/d/Projects/egorka_os/lab4/src", 128) = 35
mmap(NULL, 16432, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f84073dd000
mmap(0x7f84073de000, 4096, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1000) =
0x7f84073de000
mmap(0x7f84073df000, 4096, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x7f84073df000

```

```

mmap(0x7f84073e0000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) =
0x7f84073e0000
close(3) = 0
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=22187, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 22187, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f84071aa000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=940560, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 942344, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f84070c3000
mmap(0x7f84070d1000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xe000) =
0x7f84070d1000
mmap(0x7f840714d000, 372736, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x8a000) = 0x7f840714d000
mmap(0x7f84071a8000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xe4000) =
0x7f84071a8000
close(3) = 0
mprotect(0x7f84071a8000, 4096, PROT_READ) = 0
mprotect(0x7f84073e0000, 4096, PROT_READ) = 0
munmap(0x7f84071aa000, 22187) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x7), ...}, AT_EMPTY_PATH) =
0
write(1, "Input program code:\n", 20) = 20
write(1, " 0 -> Library switch\n", 21) = 21
write(1, " 1 -> PrimeCount\n", 17) = 17
write(1, " 2 -> Pi\n", 9) = 9
write(1, "-1 -> Exit\n", 11) = 11
newfstatat(0, "", {st_mode=S_IFIFO|0600, st_size=0, ...}, AT_EMPTY_PATH) =
0
read(0, "\n1 10\n2\n100\n0\n1\n1 10\n2\n100\n0\n2\n"..., 4096) = 38
write(1, "Enter A and B: PrimeCount(1, 10)"..., 37) = 37
write(1, "Implementation used: ./libImpl1."..., 35) =
35
write(1, "Input program code:\n", 20) = 20
write(1, " 0 -> Library switch\n", 21) = 21
write(1, " 1 -> PrimeCount\n", 17) = 17
write(1, " 2 -> Pi\n", 9) = 9
write(1, "-1 -> Exit\n", 11) = 11
write(1, "Enter K: Pi(100) = 3.131593\n", 28) =
28
write(1, "Implementation used: ./libImpl1."..., 35) =
35
write(1, "Input program code:\n", 20) = 20
write(1, " 0 -> Library switch\n", 21) = 21
write(1, " 1 -> PrimeCount\n", 17) = 17
write(1, " 2 -> Pi\n", 9) = 9
write(1, "-1 -> Exit\n", 11) = 11
munmap(0x7f84073dd000, 16432) = 0
munmap(0x7f84070c3000, 942344) = 0
openat(AT_FDCWD, "./libImpl2.so", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0777, st_size=15672, ...}, AT_EMPTY_PATH) = 0
getcwd("/mnt/d/Projects/egorka_os/lab4/src", 128) = 35
mmap(NULL, 16432, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f84073dd000
mmap(0x7f84073de000, 4096, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1000) =
0x7f84073de000
mmap(0x7f84073df000, 4096, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x7f84073df000
mmap(0x7f84073e0000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) =
0x7f84073e0000

```



```

close(3) = 0
mprotect(0x7f84073e0000, 4096, PROT_READ) = 0
write(1, "Library switched successfully!\n", 31) =
31 write(1, "Current lib: ./libImpl2.so\n", 27) = 27
write(1, "Input program code:\n", 20) = 20
write(1, " 0 -> Library switch\n", 21) = 21
write(1, " 1 -> PrimeCount\n", 17) = 17
write(1, " 2 -> Pi\n", 9) = 9
write(1, "-1 -> Exit\n", 11) = 11
write(1, "Enter A and B: PrimeCount(1, 10)"..., 37) =
37 write(1, "Implementation used: ./libImpl2."..., 35)
= 35 write(1, "Input program code:\n", 20) = 20
write(1, " 0 -> Library switch\n", 21) = 21
write(1, " 1 -> PrimeCount\n", 17) = 17
write(1, " 2 -> Pi\n", 9) = 9
write(1, "-1 -> Exit\n", 11) = 11
write(1, "Enter K: Pi(100) = 3.133787\n", 28) =
28
write(1, "Implementation used: ./libImpl2."..., 35) =
35 write(1, "Input program code:\n", 20) = 20
write(1, " 0 -> Library switch\n", 21) = 21
write(1, " 1 -> PrimeCount\n", 17) = 17
write(1, " 2 -> Pi\n", 9) = 9
write(1, "-1 -> Exit\n", 11) = 11
munmap(0x7f84073dd000, 16432) = 0
openat(AT_FDCWD, "./libImpl1.so", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0777, st_size=15648, ...}, AT_EMPTY_PATH) = 0
getcwd("/mnt/d/Projects/egorka_os/lab4/src", 128) = 35
mmap(NULL, 16432, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f84073dd000
0x7f84073de000 mmap(0x7f84073de000, 4096, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1000) =
mmap(0x7f84073df000, 4096, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x7f84073df000
0x7f84073e0000 mmap(0x7f84073e0000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) =
close(3) = 0
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=22187, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 22187, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f84071aa000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=940560, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 942344, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f84070c3000
0x7f84070d1000 mmap(0x7f84070d1000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xe000) =
mmap(0x7f840714d000, 372736, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x8a000) = 0x7f840714d000
0x7f84071a8000 mmap(0x7f84071a8000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xe4000) =
close(3) = 0
mprotect(0x7f84071a8000, 4096, PROT_READ) = 0
mprotect(0x7f84073e0000, 4096, PROT_READ) = 0
munmap(0x7f84071aa000, 22187) = 0
write(1, "Library switched successfully!\n", 31) =
31 write(1, "Current lib: ./libImpl1.so\n", 27) = 27
write(1, "Input program code:\n", 20) = 20

```

```

write(1, " 0 -> Library switch\n", 21) = 21
write(1, " 1 -> PrimeCount\n", 17)      = 17
write(1, " 2 -> Pi\n", 9)                = 9
write(1, "-1 -> Exit\n", 11)             = 11

write(1, "Enter K: Pi(100) = 3.131593\n", 28) = 28
write(1, "Implementation used: ./libImpl1."..., 35) =
35 write(1, "Input program code:\n", 20) = 20
write(1, " 0 -> Library switch\n", 21) = 21
write(1, " 1 -> PrimeCount\n", 17)      = 17
write(1, " 2 -> Pi\n", 9)                = 9
write(1, "-1 -> Exit\n", 11)             = 11
read(0, "", 4096)                        = 0
munmap(0x7f84073dd000, 16432)             = 0
munmap(0x7f84070c3000, 942344)           = 0
exit_group(0)                            = ?
+++ exited with 0 +++

```

Лабораторная работа №5-7

```

...
openat(AT_FDCWD, "/lib/libnss_db-2.35.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/libnss_db-2.35.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or
directory) munmap(0xffff84b42000, 13859) = 0
openat(AT_FDCWD, "/etc/protocols", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=2932, ...}, AT_EMPTY_PATH) =

```

```

lseek(3, 0, SEEK_SET)          = 0

read(3, "# Internet (IP) protocols\n#\n# Up"..., 4096) =
2932 read(3, "", 4096)        = 0

close(3)                       = 0

eventfd2(0, EFD_CLOEXEC)      = 3

fcntl(3, F_GETFL)              = 0x2 (flags
O_RDWR) fcntl(3, F_SETFL,
O_RDWR|O_NONBLOCK) = 0

fcntl(3, F_GETFL)              = 0x802 (flags
O_RDWR|O_NONBLOCK) fcntl(3, F_SETFL,
O_RDWR|O_NONBLOCK) = 0

getpid()                       = 5151

getpid()                       = 5151

...
5153 strace: Process 5153 attached

[pid 5151] clock_nanosleep(CLOCK_REALTIME, 0, {tv_sec=1, tv_nsec=0}, <unfinished ...>

[pid 5153] set_robust_list(0xffff84b0c100, 24) = 0

[pid 5153] eventfd2(0, EFD_CLOEXEC) = 4

[pid 5153] fcntl(4, F_GETFL)      = 0x2 (flags O_RDWR)

[pid 5153] fcntl(4, F_SETFL, O_RDWR|O_NONBLOCK) = 0

[pid 5153] fcntl(4, F_GETFL)      = 0x802 (flags
O_RDWR|O_NONBLOCK) [pid 5153] fcntl(4, F_SETFL,
O_RDWR|O_NONBLOCK) = 0

[pid 5153] getpid()              = 5153

[pid 5153] getpid()              = 5153

[pid 5153] eventfd2(0, EFD_CLOEXEC) = 5

[pid 5153] fcntl(5, F_GETFL)      = 0x2 (flags O_RDWR)

[pid 5153] fcntl(5, F_SETFL, O_RDWR|O_NONBLOCK) = 0

[pid 5153] fcntl(5, F_GETFL)      = 0x802 (flags
O_RDWR|O_NONBLOCK) [pid 5153] fcntl(5, F_SETFL,
O_RDWR|O_NONBLOCK) = 0

[pid 5153] getpid()              = 5153

[pid 5153] epoll_create1(E POLL_CLOEXEC) = 6

```

```

[pid 5153] epoll_ctl(6, EPOLL_CTL_ADD, 5, {events=0, data={u32=3664091408, u64=187650785253648}}) = 0

[pid 5153] epoll_ctl(6, EPOLL_CTL_MOD, 5, {events=EPOLLIN, data={u32=3664091408, u64=187650785253648}}) = 0

[pid 5153] getpid()                = 5153

[pid 5153] rt_sigaction(SIGRT_1, {sa_handler=0xffff8493a700, sa_mask=[], sa_flags=SA_ONSTACK|SA_RESTART|SA_SIGINFO}, NULL, 8) = 0

[pid 5153] rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0

[pid 5153] mmap(NULL, 8454144, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0xffff83800000

[pid 5153] mprotect(0xffff83810000, 8388608, PROT_READ|PROT_WRITE) = 0

[pid 5153] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

[pid 5153] clone(child_stack=0xffff8400e6e0,
flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID,
strace: Process 5154 attached

<unfinished ...>

[pid 5154] rseq(0xffff8400f640, 0x20, 0, 0xd428bc00 <unfinished ...>

[pid 5153] <... clone resumed>, parent_tid=[5154], tls=0xffff8400f660, child_tidptr=0xffff8400ef70) = 5154

[pid 5154] <... rseq resumed>      = 0

[pid 5153] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 5154] set_robust_list(0xffff8400ef80, 24 <unfinished ...>

[pid 5153] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 5154] <... set_robust_list resumed>) = 0

[pid 5153] eventfd2(0, EFD_CLOEXEC <unfinished ...>

[pid 5154] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 5153] <... eventfd2 resumed>) = 7

[pid 5154] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 5153] fcntl(7, F_GETFL <unfinished ...>

[pid 5154] rt_sigprocmask(SIG_BLOCK, ~[RTMIN RT_1], <unfinished ...>

[pid 5153] <... fcntl resumed>    = 0x2 (flags O_RDWR)

[pid 5154] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 5153] fcntl(7, F_SETFL, O_RDWR|O_NONBLOCK <unfinished ...>

[pid 5154] sched_getparam(5154, <unfinished ...>

[pid 5153] <... fcntl resumed>    = 0

[pid 5154] <... sched_getparam resumed>[0] = 0

[pid 5153] fcntl(7, F_GETFL <unfinished ...>

[pid 5154] sched_getscheduler(5154 <unfinished ...>

[pid 5153] <... fcntl resumed>    = 0x802 (flags

O_RDWR|O_NONBLOCK) [pid 5154] <... sched_getscheduler resumed>) = 0

(SCHED_OTHER)

[pid 5153] fcntl(7, F_SETFL, O_RDWR|O_NONBLOCK <unfinished ...>

[pid 5154] sched_setscheduler(5154, SCHED_OTHER, [0] <unfinished ...>

[pid 5153] <... fcntl resumed>    = 0

```

[pid 5154] <... sched_setscheduler resumed>) = 0

[pid 5153] getpid(<unfinished ...>

[pid 5154] prctl(PR_SET_NAME, "ZMQbg/Reaper" <unfinished ...>

[pid 5153] <... getpid resumed>) = 5153

[pid 5154] <... prctl resumed>) = 0

[pid 5153] epoll_create1(EPOLL_CLOEXEC <unfinished ...>

[pid 5154] epoll_pwait(6, <unfinished ...>

[pid 5153] <... epoll_create1 resumed>) = 8

[pid 5153] epoll_ctl(8, EPOLL_CTL_ADD, 7, {events=0, data={u32=3664093488, u64=187650785255728}}) = 0

[pid 5153] epoll_ctl(8, EPOLL_CTL_MOD, 7, {events=EPOLLIN, data={u32=3664093488, u64=187650785255728}}) = 0

[pid 5153] mmap(NULL, 8454144, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0xffff82e00000

[pid 5153] mprotect(0xffff82e10000, 8388608, PROT_READ|PROT_WRITE) = 0

[pid 5153] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

[pid 5153] clone(child_stack=0xffff8360e6e0, flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEAR_TID|strace: Process 5155 attached

<unfinished ...>

[pid 5155] rseq(0xffff8360f640, 0x20, 0, 0xd428bc00 <unfinished ...>

[pid 5153] <... clone resumed>, parent_tid=[5155], tls=0xffff8360f660, child_tidptr=0xffff8360ef70) = 5155

[pid 5155] <... rseq resumed>) = 0

[pid 5153] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 5155] set_robust_list(0xffff8360ef80, 24 <unfinished ...>

[pid 5153] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 5155] <... set_robust_list resumed>) = 0

[pid 5153] eventfd2(0, EFD_CLOEXEC <unfinished ...>

[pid 5155] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 5153] <... eventfd2 resumed>) = 9

[pid 5155] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 5153]fcntl(9, F_GETFL <unfinished ...>

[pid 5155] rt_sigprocmask(SIG_BLOCK, ~[RTMIN RT_1], <unfinished ...>

[pid 5153] <... fcntl resumed>) = 0x2 (flags O_RDONLY)

[pid 5155] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 5153]fcntl(9, F_SETFL, O_RDONLY|O_NONBLOCK <unfinished ...>

[pid 5155] sched_getparam(5155, <unfinished ...>

[pid 5153] <... fcntl resumed>) = 0

[pid 5155] <... sched_getparam resumed>[0]) = 0

[pid 5153]fcntl(9, F_GETFL <unfinished ...>

[pid 5155] sched_getscheduler(5155 <unfinished ...>

```

[pid 5153] <... fcntl resumed>)      = 0x802 (flags
O_RDWR|O_NONBLOCK) [pid 5155] <... sched_getscheduler resumed>) = 0
(SCHED_OTHER)

[pid 5153] fcntl(9, F_SETFL, O_RDWR|O_NONBLOCK <unfinished ...>

[pid 5155] sched_setscheduler(5155, SCHED_OTHER, [0] <unfinished ...>

[pid 5153] <... fcntl resumed>)      = 0

[pid 5155] <... sched_setscheduler resumed>) = 0

[pid 5153] getpid( <unfinished ...>

[pid 5155] prctl(PR_SET_NAME, "ZMQbg/IO/0" <unfinished ...>

[pid 5153] <... getpid resumed>)      = 5153

[pid 5155] <... prctl resumed>)      =

0 [pid 5153] getpid( <unfinished ...>

[pid 5155] epoll_pwait(8, <unfinished ...>

[pid 5153] <... getpid resumed>)      =

5153

[pid 5153] ppoll([{fd=9, events=POLLIN}], 1, {tv_sec=0, tv_nsec=0}, NULL, 0) = 0 (Timeout)

[pid 5153] socket(AF_NETLINK, SOCK_RAW|SOCK_CLOEXEC, NETLINK_ROUTE) = 10

[pid 5153] bind(10, {sa_family=AF_NETLINK, nl_pid=0, nl_groups=00000000}, 12) = 0

[pid 5153] getsockname(10, {sa_family=AF_NETLINK, nl_pid=5153, nl_groups=00000000}, [12]) = 0

[pid 5153] sendto(10, [{nlmsg_len=20, nlmsg_type=RTM_GETLINK, nlmsg_flags=NLM_F_REQUEST|NLM_F_DUMP, nlmsg_seq=1741222074,
nlmsg_pid=0}, {ifl_family=AF_UNSPEC, ...}], 20, 0, {sa_family=AF_NETLINK, nl_pid=0, nl_groups=00000000}, 12) = 20

...

[pid 5153] getsockname(10, {sa_family=AF_INET, sin_port=htons(5551), sin_addr=inet_addr("127.0.0.1")}, [128 => 16]) = 0

[pid 5153] getsockname(10, {sa_family=AF_INET, sin_port=htons(5551), sin_addr=inet_addr("127.0.0.1")}, [128 => 16]) = 0

[pid 5153] getpid()                  = 5153

[pid 5153] write(7, "\1\0\0\0\0\0\0\0", 8 <unfinished ...>

[pid 5155] <... epoll_pwait resumed>[{events=EPOLLIN, data={u32=3664093488, u64=187650785255728}}, 256, -1, NULL, 8) = 1

[pid 5153] <... write resumed>)      = 8

[pid 5155] getpid( <unfinished ...>

[pid 5153] getpid( <unfinished ...>

[pid 5155] <... getpid resumed>)      =

5153 [pid 5153] <... getpid resumed>)

=

5153

[pid 5155] ppoll([{fd=7, events=POLLIN}], 1, {tv_sec=0, tv_nsec=0}, NULL, 0 <unfinished ...>

[pid 5153] write(9, "\1\0\0\0\0\0\0\0", 8 <unfinished ...>

[pid 5155] <... ppoll resumed>)      = 1 ([{fd=7, revents=POLLIN}], left {tv_sec=0,
tv_nsec=0}) [pid 5153] <... write resumed>) = 8

[pid 5155] getpid( <unfinished ...>

```

[pid 5153] write(1, "Compute Node 1 started, listenin"..., 58 <unfinished ...>

[pid 5155] <... getpid resumed> = 5153

[pid 5153] <... write resumed> =

58 [pid 5155] read(7, <unfinished ...>

[pid 5153] getpid(<unfinished ...>

[pid 5155] <... read resumed>"\1\0\0\0\0\0\0", 8) = 8

```

[pid 5153] <... getpid resumed>      = 5153

[pid 5155] mmap(NULL, 134217728, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0 <unfinished ...>

[pid 5153] ppoll([{fd=9, events=POLLIN}], 1, NULL, NULL, 0 <unfinished ...>

[pid 5155] <... mmap resumed>        = 0xffff7ae00000

[pid 5153] <... ppoll resumed>       = 1 ([{fd=9,
revents=POLLIN}]) [pid 5155] munmap(0xffff7ae00000, 18874368
<unfinished ...>

[pid 5153] getpid( <unfinished ...>

[pid 5155] <... munmap resumed>      = 0

[pid 5153] <... getpid resumed>      =
5153

[pid 5155] munmap(0xffff80000000, 48234496 <unfinished ...>

[pid 5153] read(9, <unfinished ...>

[pid 5155] <... munmap resumed>      = 0

[pid 5153] <... read resumed>"\1\0\0\0\0\0\0", 8) = 8

[pid 5155] mprotect(0xffff7c000000, 135168, PROT_READ|PROT_WRITE <unfinished ...>

[pid 5153] getpid( <unfinished ...>

[pid 5155] <... mprotect resumed>) = 0 [pid
5153] <... getpid resumed>          =
5153

[pid 5155] epoll_ctl(8, EPOLL_CTL_ADD, 10, {events=0, data={u32=2080377712, u64=281472762121072}} <unfinished ...>

[pid 5153] ppoll([{fd=9, events=POLLIN}], 1, {tv_sec=0, tv_nsec=0}, NULL, 0 <unfinished ...>

[pid 5155] <... epoll_ctl resumed>) = 0

[pid 5153] <... ppoll resumed>       = 0 (Timeout)

[pid 5155] epoll_ctl(8, EPOLL_CTL_MOD, 10, {events=EPOLLIN, data={u32=2080377712, u64=281472762121072}} <unfinished ...>

[pid 5153] getpid( <unfinished ...>

[pid 5155] <... epoll_ctl resumed>) = 0 [pid
5153] <... getpid resumed>          =
5153 [pid 5155] getpid( <unfinished ...>

[pid 5153] ppoll([{fd=9, events=POLLIN}], 1, NULL, NULL, 0 <unfinished ...>

[pid 5155] <... getpid resumed>      = 5153

[pid 5155] ppoll([{fd=7, events=POLLIN}], 1, {tv_sec=0, tv_nsec=0}, NULL, 0) = 0 (Timeout)

[pid 5155] epoll_pwait(8, <unfinished ...>

[pid 5151] <... clock_nanosleep resumed>0xffffe4f67958) = 0

[pid 5151] eventfd2(0, EFD_CLOEXEC) = 4

[pid 5151] fcntl(4, F_GETFL)        = 0x2 (flags O_RDWR)

[pid 5151] fcntl(4, F_SETFL, O_RDWR|O_NONBLOCK) = 0

[pid 5151] fcntl(4, F_GETFL)        = 0x802 (flags
O_RDWR|O_NONBLOCK) [pid 5151] fcntl(4, F_SETFL,

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O_RDWR|O_NONBLOCK) = 0

[pid 5151] getpid() = 515

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[pid 5151] epoll_create1(EPOLL_CLOEXEC) = 5

[pid 5151] epoll_ctl(5, EPOLL_CTL_ADD, 4, {events=0, data={u32=3664068656, u64=187650785230896}}) = 0

[pid 5151] epoll_ctl(5, EPOLL_CTL_MOD, 4, {events=EPOLLIN, data={u32=3664068656, u64=187650785230896}}) = 0

[pid 5151] getpid() = 5151

[pid 5151] rt_sigaction(SIGRT_1, {sa_handler=0xffff8493a700, sa_mask=[], sa_flags=SA_ONSTACK|SA_RESTART|SA_SIGINFO}, NULL, 8) = 0

[pid 5151] rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0

[pid 5151] mmap(NULL, 8454144, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0xffff83800000

[pid 5151] mprotect(0xffff83810000, 8388608, PROT_READ|PROT_WRITE) = 0

[pid 5151] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

[pid 5151] clone(child_stack=0xffff8400e6e0,
flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, parent_tid=[5172], tls=0xffff8400f660, child_tidptr=0xffff8400ef70) =

5172 strace: Process 5172 attached

[pid 5151] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 5172] rseq(0xffff8400f640, 0x20, 0, 0xd428bc00 <unfinished ...>

[pid 5151] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 5172] <... rseq resumed> = 0

[pid 5151] eventfd2(0, EFD_CLOEXEC <unfinished ...>

[pid 5172] set_robust_list(0xffff8400ef80, 24 <unfinished ...>

[pid 5151] <... eventfd2 resumed>) = 6

[pid 5172] <... set_robust_list resumed>) = 0

[pid 5151] fcntl(6, F_GETFL <unfinished ...>

[pid 5172] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 5151] <... fcntl resumed> = 0x2 (flags O_RDWR)

[pid 5172] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 5151] fcntl(6, F_SETFL, O_RDWR|O_NONBLOCK <unfinished ...>

[pid 5172] rt_sigprocmask(SIG_BLOCK, ~[RTMIN RT_1], <unfinished ...>

[pid 5151] <... fcntl resumed> = 0

[pid 5172] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 5151] fcntl(6, F_GETFL <unfinished ...>

[pid 5172] sched_getparam(5172, <unfinished ...>

[pid 5151] <... fcntl resumed> = 0x802 (flags
O_RDWR|O_NONBLOCK) [pid 5172] <... sched_getparam resumed>[0]) = 0

[pid 5151] fcntl(6, F_SETFL, O_RDWR|O_NONBLOCK <unfinished ...>

[pid 5172] sched_getscheduler(5172 <unfinished ...>

[pid 5151] <... fcntl resumed> = 0

[pid 5172] <... sched_getscheduler resumed>) = 0 (SCHED_OTHER)

[pid 5151] getpid( <unfinished ...>

```

```
[pid 5172] sched_setscheduler(5172, SCHED_OTHER, [0] <unfinished ...>

[pid 5151] <... getpid resumed>      = 5151

[pid 5172] <... sched_setscheduler resumed> = 0

[pid 5151] epoll_create1(EPoll_CLOEXEC <unfinished ...>

[pid 5172] prctl(PR_SET_NAME, "ZMQbg/Reaper" <unfinished ...>

[pid 5151] <... epoll_create1 resumed> = 7

[pid 5172] <... prctl resumed>      = 0

[pid 5151] epoll_ctl(7, EPOLL_CTL_ADD, 6, {events=0, data={u32=3664090192, u64=187650785252432}} <unfinished ...>

[pid 5172] epoll_pwait(5, <unfinished ...>

[pid 5151] <... epoll_ctl resumed> = 0

[pid 5151] epoll_ctl(7, EPOLL_CTL_MOD, 6, {events=EPOLLIN, data={u32=3664090192, u64=187650785252432}}) = 0

[pid 5151] mmap(NULL, 8454144, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0xffff82e00000

[pid 5151] mprotect(0xffff82e10000, 8388608, PROT_READ|PROT_WRITE) = 0

[pid 5151] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

[pid 5151] clone(child_stack=0xffff8360e6e0,
flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, parent_tid=[5173], tls=0xffff8360f660, child_tidptr=0xffff8360ef70) =

5173 strace: Process 5173 attached

[pid 5151] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 5173] rseq(0xffff8360f640, 0x20, 0, 0xd428bc00 <unfinished ...>

[pid 5151] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 5173] <... rseq resumed>      = 0

[pid 5151] eventfd2(0, EFD_CLOEXEC) = 8

[pid 5173] set_robust_list(0xffff8360ef80, 24 <unfinished ...>

[pid 5151] fcntl(8, F_GETFL <unfinished ...>

[pid 5173] <... set_robust_list resumed> = 0

[pid 5151] <... fcntl resumed>      = 0x2 (flags O_RDWR)

[pid 5173] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 5151] fcntl(8, F_SETFL, O_RDWR|O_NONBLOCK <unfinished

...

```